

CARBON AND ITS COMPOUND MCQ

1. Organic compounds are defined as those containing carbon covalently linked primarily to:

- a) metals
- b) hydrogen, oxygen, nitrogen
- c) halogens only
- d) noble gases?

Correct answer: b

2. Carbon compounds exist in our food, clothes, and pencil lead because carbon forms:

- a) ionic bonds only
- b) covalent bonds with many elements
- c) metallic bonds
- d) coordinate bonds?

Correct answer: b

3. A key characteristic of organic compounds distinguishing them from inorganic is:

- a) high reactivity
- b) isomerism phenomenon
- c) high melting point
- d) water solubility?

Correct answer: b

4. Butane C_4H_{10} shows isomerism with n-butane and:

- a) propane
- b) iso-butane
- c) ethane

d) pentane?

Correct answer: b

5. Organic compounds are generally insoluble in water but soluble in:

a) acids

b) organic solvents like ether

c) bases

d) salts?

Correct answer: b

6. Acyclic compounds have carbon atoms arranged in:

a) rings

b) linear or branched chains

c) only straight chains

d) aromatic rings?

Correct answer: b

7. Saturated acyclic compounds contain only:

a) double bonds

b) triple bonds

c) single bonds between carbons

d) multiple bonds?

Correct answer: c

8. Unsaturated compounds have at least one:

a) single C-C bond

b) double or triple C-C bond

c) ionic bond

d) coordinate bond?

Correct answer: b

9. Heterocyclic compounds contain carbon ring with:

- a) only carbon atoms
- b) other atoms like O, N, S
- c) only hydrogen
- d) metals?

Correct answer: b

10. Aromatic compounds contain:

- a) only single bonds
- b) benzene rings with alternate double bonds
- c) only triple bonds
- d) no rings?

Correct answer: b

11. Hydrocarbons consist exclusively of:

- a) C and O
- b) C and H only
- c) C, H, N
- d) C and halogens?

Correct answer: b

12. Alkanes general formula is:

- a) C_nH_{2n}
- b) C_nH_{2n+2}
- c) C_nH_{2n-2}
- d) C_nH_n ?

Correct answer: b

13. Alkenes contain at least one:

- a) C-C single bond

b) C=C double bond

c) C $\equiv$ C triple bond

d) C-O bond?

Correct answer: b

14. Simplest alkyne is:

a) methane

b) ethyne C₂H₂

c) ethene

d) propane?

Correct answer: b

15. Methene and methyne do not exist because carbon needs:

a) one H

b) multiple bonds requiring more carbons

c) ionic structure

d) ring formation?

Correct answer: b

16. Bromine water test decolourises with:

a) saturated compounds

b) unsaturated compounds

c) alkanes only

d) aromatics?

Correct answer: b

17. Functional group gives compound its:

a) physical state

b) characteristic chemical properties

c) colour

d) odour?

Correct answer: b

18. Homologous series members differ by:

a) -CH_3

b) -CH_2 group

c) -OH

d) =O ?

Correct answer: b

19. In homologous series, physical properties show:

a) irregular variation

b) regular gradation with molecular mass

c) no change

d) sudden jumps?

Correct answer: b

20. IUPAC name components are root word, prefix, and:

a) isotope

b) suffix

c) radical

d) ion?

Correct answer: b

21. Root word indicates number of:

a) hydrogens

b) carbon atoms in parent chain

c) oxygens

d) bonds?

Correct answer: b

22. For pentane chain, root word is:

- a) pent
- b) penta
- c) pentane
- d) ane?

Correct answer: a

23. Primary suffix 'ane' indicates:

- a) double bond
- b) all single bonds
- c) triple bond
- d) ring?

Correct answer: b

24. In naming, chain numbered from end giving lowest number to:

- a) hydrogens
- b) substituent or functional group
- c) carbons
- d) oxygens?

Correct answer: b

25. For pentene with double bond at position 1:

- a) pent-1-ene
- b) 1-pentene
- c) pentene
- d) pent-2-ene?

Correct answer: a

26. Ethanol common name is:

- a) methyl alcohol

- b) ethyl alcohol
- c) wood spirit
- d) rubbing alcohol?

Correct answer: b

27. Ethanol manufactured by fermentation of:

- a) starch
- b) molasses containing sucrose
- c) glucose directly
- d) proteins?

Correct answer: b

28. Fermentation requires yeast enzymes:

- a) invertase and zymase
- b) amylase only
- c) lipase
- d) protease?

Correct answer: a

29. Rectified spirit is:

- a) 100% ethanol
- b) 95.5% ethanol + 4.5% water
- c) 50% ethanol
- d) pure water?

Correct answer: b

30. Absolute alcohol obtained by refluxing rectified spirit over:

- a) H_2SO_4
- b) quicklime
- c) NaOH

d) KMnO_4 ?

Correct answer: b

31. Ethanol boiling point is:

a) 184K like ethane

b) 351K

c) 273K

d) 373K?

Correct answer: b

32. Dehydration of ethanol at 443K with conc H_2SO_4 gives:

a) ethene

b) ethane

c) ethyne

d) ethanal?

Correct answer: a

33. Ethanol + Na gives:

a) H_2 and sodium ethoxide

b) O_2

c) CO_2

d) ethene?

Correct answer: a

34. Oxidation of ethanol with KMnO_4 gives:

a) ethanal

b) ethanoic acid

c) ethene

d) hydrogen?

Correct answer: b

35.Esterification: ethanol + ethanoic acid gives:

- a) ethyl ethanoate
- b) ethanal
- c) ethene
- d) sodium ethanoate?

Correct answer: a

36.Dehydrogenation of ethanol vapour over Cu at 573K gives:

- a) ethene
- b) acetaldehyde
- c) CO₂
- d) H₂O?

Correct answer: b

37.Ethanol uses as antiseptic in:

- a) food
- b) medical wipes
- c) fuels only
- d) paints?

Correct answer: b

38.Ethanoic acid also called:

- a) formic acid
- b) acetic acid
- c) propanoic
- d) butyric acid?

Correct answer: b

39.Glacial acetic acid refers to pure ethanoic acid because it forms:

- a) solid ice-like flakes on cooling

- b) gas
- c) crystals in water
- d) vapour?

Correct answer: a

40. Ethanoic acid reacts with Na to give:

- a) H₂ and sodium ethanoate
- b) CO₂
- c) O₂
- d) ethane?

Correct answer: a

41. Reaction with NaHCO₃ gives ethanoic acid:

- a) no reaction
- b) CO₂ effervescence
- c) H₂
- d) O₂?

Correct answer: b

42. Decarboxylation of sodium ethanoate with soda lime gives:

- a) ethane
- b) methane
- c) ethene
- d) CO₂?

Correct answer: b

43. Vinegar contains ethanoic acid concentration of:

- a) 100%
- b) lower concentrations as preservative
- c) 50%

d) gas form?

Correct answer: b

44. In daily life, hydrocarbons used in:

a) LPG, petrol

b) sugars

c) proteins

d) vitamins?

Correct answer: a

45. Aldehydes like formaldehyde used in:

a) formalin for preservation

b) fuels

c) soaps

d) detergents?

Correct answer: a

46. Acetone ketone used as:

a) food additive

b) nail polish remover

c) antiseptic

d) explosive?

Correct answer: b

47. Soaps are sodium/potassium salts of:

a) sulfonic acids

b) long chain carboxylic acids

c) alcohols

d) hydrocarbons?

Correct answer: b

48. Hard soaps made using:

- a) KOH
- b) NaOH
- c) NH₃
- d) Ca(OH)₂?

Correct answer: b

49. Kettle process for soap involves:

- a) saponification then salting out
- b) distillation
- c) fermentation
- d) oxidation?

Correct answer: a

50. Hard water forms scum with soap due to:

- a) Na⁺, K⁺
- b) Ca²⁺, Mg²⁺ precipitates
- c) Cl⁻ ions
- d) SO₄²⁻?

Correct answer: b

51. Soap molecule has hydrophilic head and:

- a) hydrophilic tail
- b) hydrophobic hydrocarbon tail
- c) ionic tail
- d) polar tail?

Correct answer: b

52. Micelles form when soap dissolves in water trapping:

- a) water

b) dirt and oil

c) minerals

d) bacteria?

Correct answer: b

53. Detergents are salts of:

a) carboxylic acids

b) sulfonic acids

c) alcohols

d) ketones?

Correct answer: b

54. Detergents work in hard water unlike soap because they don't form:

a) micelles

b) precipitates with Ca^{2+} , Mg^{2+}

c) foam

d) suds?

Correct answer: b

55. Biodegradable detergents have:

a) branched chains

b) straight hydrocarbon chains

c) rings

d) aromatics?

Correct answer: b

56. TFM in soap stands for and higher TFM means:

a) total fatty matter, better quality

b) thin film material

c) thermal fusion mass

d) toxic free matter?

Correct answer: a

57. Alicyclic compounds are:

a) aromatic

b) saturated/unsaturated carbocyclic rings

c) heterocyclic

d) acyclic?

Correct answer: b

58. Carbocyclic means ring contains:

a) heteroatoms

b) only carbon atoms

c) hydrogen only

d) oxygen?

Correct answer: b

59. Lower hydrocarbons like methane are:

a) solids

b) liquids

c) gases at room temperature

d) plasmas?

Correct answer: c

60. Boiling point of hydrocarbons increases with:

a) decrease in carbon atoms

b) increase in carbon atoms

c) double bonds

d) triple bonds?

Correct answer: b

61. Most reactive hydrocarbons due to triple bond:

- a) alkanes
- b) alkenes
- c) alkynes
- d) aromatics?

Correct answer: c

62. Propan-1-ol named by numbering from OH group getting:

- a) highest number
- b) lowest locant 1
- c) middle number
- d) no number?

Correct answer: b

63. Ethanoic acid suffix is:

- a) -al
- b) -one
- c) oic acid
- d) -ol?

Correct answer: c

64. Butan-2-one has carbonyl at position:

- a) 1
- b) 2
- c) 3
- d) 4?

Correct answer: b

65. 1-chlorobutane prefix indicates chloro at:

- a) carbon 4

- b) carbon 1
- c) carbon 2
- d) no position?

Correct answer: b

66. Fermentation temperature maintained at:

- a) 273K
- b) 303K
- c) 443K
- d) 573K?

Correct answer: b

67. Ethanol completely miscible with:

- a) oil
- b) benzene
- c) water any proportion
- d) mercury?

Correct answer: c

68. Colour change in ethanol oxidation: orange $K_2Cr_2O_7$ to:

- a) red
- b) green
- c) blue
- d) yellow?

Correct answer: b

69. Ester has:

- a) sour taste
- b) fruity odour
- c) burning taste

d) pungent smell?

Correct answer: b

70. Ethanoic acid higher boiling point than:

a) alkanes

b) alcohols, aldehydes

c) ketones only

d) esters?

Correct answer: b

71. Soda lime for decarboxylation is:

a) NaOH + CaO

b) NaCl + CaO

c) Na₂CO₃

d) KOH?

Correct answer: a

72. In IUPAC, multiple identical substituents use:

a) di, tri prefixes

b) single prefix

c) no locants

d) different names?

Correct answer: a

73. For branched chain, parent is:

a) shortest chain

b) longest continuous chain

c) ring

d) functional group chain?

Correct answer: b

74.Secondary suffix for aldehyde:

- a) -oic acid
- b) -al
- c) -one
- d) -ol?

Correct answer: b

75.Washing machine protected by sodium silicate in:

- a) soap
- b) detergent
- c) ethanol
- d) vinegar?

Correct answer: b

76.Fluorescent in detergent acts as:

- a) corrosion inhibitor
- b) whitening agent
- c) anti-caking
- d) enzyme?

Correct answer: b

77.Oxygen bleaches like sodium perborate remove:

- a) oil stains
- b) certain cloth stains
- c) grease
- d) all dirt?

Correct answer: b

78.Detergents better for:

- a) soft water only

b) hard and acidic water

c) only saline water

d) distilled water?

Correct answer: b

79. Soft soaps used for:

a) washing clothes

b) body cleansing

c) dishes

d) cars?

Correct answer: b

80. Van der Waals forces in alkanes increase with:

a) molecular mass

b) double bonds

c) functional groups

d) rings?

Correct answer: a

81. Chemical properties same in homologous series due to:

a) same functional group

b) different chains

c) -CH₂ difference

d) molecular mass?

Correct answer: a

82. Aldehydes prepared by oxidising:

a) secondary alcohols

b) primary alcohols

c) tertiary alcohols

d) carboxylic acids?

Correct answer: b

83. Benzaldehyde used in:

a) fuels

b) perfumes, almond flavour

c) soaps

d) detergents?

Correct answer: b

84. Cyclohexanone used in:

a) nylon production

b) food

c) medicine

d) explosives?

Correct answer: a

85. Diethyl ether with ethanol is:

a) spirit of ether painkiller

b) soap

c) detergent

d) ester?

Correct answer: a

86. Esters in cooking oils are:

a) hydrocarbons

b) lipids

c) alcohols

d) ketones?

Correct answer: b

87. Nitro-glycerine ester used as:

- a) flavour
- b) explosive
- c) solvent
- d) antiseptic?

Correct answer: b

88. Bureau Indian Standards TFM for bathing bar minimum:

- a) 76%
- b) 60% not less than 40%
- c) 50%
- d) 90%?

Correct answer: b

89. Higher TFM soap provides:

- a) more lather
- b) moisturizing properties
- c) hardness
- d) scum?

Correct answer: b

90. Non-biodegradable detergents have:

- a) straight chains
- b) highly branched chains
- c) short chains
- d) rings?

Correct answer: b

91. In combustion, hydrocarbons produce:

- a) $\text{CO} + \text{H}_2\text{O}$

b) $\text{CO}_2 + \text{H}_2\text{O}$

c) $\text{C} + \text{H}_2$

d) CH_4 ?

Correct answer: b

92. Test for unsaturation uses:

a) KMnO_4

b) bromine water decolourisation

c) Na

d) H_2SO_4 ?

Correct answer: b

93. IUPAC for $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$:

a) ethanol

b) propan-1-ol

c) propanol

d) 1-propanol?

Correct answer: b

94. For CH_3COCH_3 name is:

a) propanone

b) propanal

c) propanoic acid

d) 2-propanol?

Correct answer: a

95. Yeast added after:

a) distillation

b) nitrogen source in fermentation

c) salting out

d) oxidation?

Correct answer: b

96. Wash in ethanol production is:

a) pure ethanol

b) fermented liquid 15-18% alcohol

c) molasses

d) absolute alcohol?

Correct answer: b

97. Power alcohol is mixture of:

a) ethanol + methanol

b) petrol + ethanol

c) water + ethanol

d) NaOH + ethanol?

Correct answer: b

98. Denatured spirit has ethanol +:

a) water

b) pyridine

c) sugar

d) salt?

Correct answer: b

99. Ethers used as:

a) fuels

b) anaesthetic agents

c) preservatives

d) bleaches?

Correct answer: b

100. Surfactants in detergents from:

- a) fatty acids
- b) petroleum hydrocarbons + H_2SO_4
- c) molasses
- d) lime?

Correct answer: b